

Motion and Shifts as Weapons

Why NFL offenses now move someone before most snaps, what each kind of motion is for, and how to read the defense's answer before the ball is snapped.

First-and-10, early in the second quarter. The receiver split wide on the right turns and jogs toward the ball, then breaks into a sprint, running flat along the line, across the whole formation. The cornerback who had been standing seven yards in front of him turns and runs too, step for step, all the way across the field. "There's your man coverage," says the analyst, and before he finishes the sentence the ball is snapped, the receiver takes a little flip from the quarterback without breaking stride, and he is ten yards upfield before the corner chasing him can turn the corner. How did the analyst know? Why did the offense want him to know? And if the run was coming anyway, why didn't the defense simply let the receiver go? This chapter is about the most common thing an NFL offense does before the snap, moving somebody, and about what each kind of movement is for.

You'll be able to...

- Name the common motions on sight: **jet**, **fly**, **rocket**, **orbit**, **return (yo-yo)**, **zoom**, **glide**, **short (rip/liz)**, the tight end **trade** and the **shift to bunch**, and say when a motion is happening at the **snap**.
- Explain the five jobs motion does: it asks *man or zone?*, changes the formation's strength and numbers, creates leverage, gives a player a running start, and stresses the defense's communication.
- Read the defense's answer as it happens (a defender **travels**, the defense **bumps**, or the safeties **spin**) and turn it into a coverage guess, while knowing when that guess can be wrong.
- Explain how the NFL's motion rule differs from college and Canadian football, and what the league changed in 2024 after the "cheat motion".
- Describe the 2016–2025 motion boom with numbers, and explain why PFF, FTN, ESPN and Next Gen Stats publish such different motion rates.
- Run the **watching checklist so far**: situation → personnel → formation and strength → the motion response → box → shell.

i You'll need

- **The pre-snap rules** (The Pre-Snap Phase): after a shift (two or more players moving at once) everyone must be set for a full second; in motion only one backfield player may be moving at the snap, parallel to or away from the line, never toward it.
- **Formations** (Spread and Modern Formations): 2x2 is two receivers on each side, 3x1 (trips) three to one side and one to the other, a bunch is three receivers squeezed into a tight triangle, and an empty set has no back beside the quarterback. The side with more players (usually the tight end's side) is the strength.
- **A first look at defense** (A First Look at Defense): in man coverage each defender is assigned a receiver; in zone coverage each is assigned an area. The safeties' picture is one-high (one deep in the middle) or two-high (two deep, one per side), and the box is the area near the ball where run defenders stand.

Motion has a job

The Pre-Snap Phase taught the rules: who may move, in which direction, and when everyone has to freeze. Those rules are why the movement you see looks the way it does. A receiver running across the formation moves *flat*, parallel to the line, because he is not allowed to be angling toward it when the ball is snapped. Two receivers never sprint at the same moment at the snap, because only one player may be moving. The rules set the shape. This chapter is about the *purpose*.

Here is the idea to hold on to: **no good offense moves a player for decoration**. Kyle Shanahan, whose 49ers have been among the league's heaviest motion users since 2017, put it plainly: "You do it for a reason. You do it to help one of your players or you do it because you see something that you think will affect the defense." Sean McVay of the Rams added the warning every coach repeats: "It's not just to look cool."¹

¹Nick Wagoner and Lindsey Thiry, "49ers' Kyle Shanahan and Rams' Sean McVay always thinking pre-snap motion," ESPN (Jan. 29, 2022): 49ers 69.1% pre-snap motion 2017–2021 (first), Rams 46.3% (eighth); 2021: 49ers 76.9% (first), Rams 47.4%; league 34.9% (2017) to 46.8% (2021); quotes from Shanahan and McVay: https://www.espn.com/nfl/story/_/id/32598845/

Every motion does one or more of five jobs:

1. **It asks a question: man or zone?** Someone on the defense has to deal with the moving player, and *how* they deal with him is a clue to the coverage.
2. **It changes the strength and the numbers.** A player who moves from one side to the other changes how many receivers are on each side, and the defense has to rebalance in a second or two.
3. **It creates leverage.** A defender's position relative to his receiver, inside or outside, in front or behind, is planned against a *standing* receiver. A moving one changes it.
4. **It gives a running start.** A player moving at the snap is already at speed when the play begins, and nobody on the defense is.
5. **It stresses communication.** Each time the formation changes, the defense has to re-declare its calls, and every re-declared call is a chance for two defenders to take the same man and none to take another.

And every motion has a cost: it spends time on the play clock, it has to be timed precisely, and it can tip the play if the same motion always comes before the same play. The rest of the chapter takes the jobs one at a time. First, the vocabulary: the motions themselves.

The motion family

Motions are named by **path** (where the player goes) and **timing** (whether he is still moving when the ball is snapped). Here is the family the rest of this book uses.

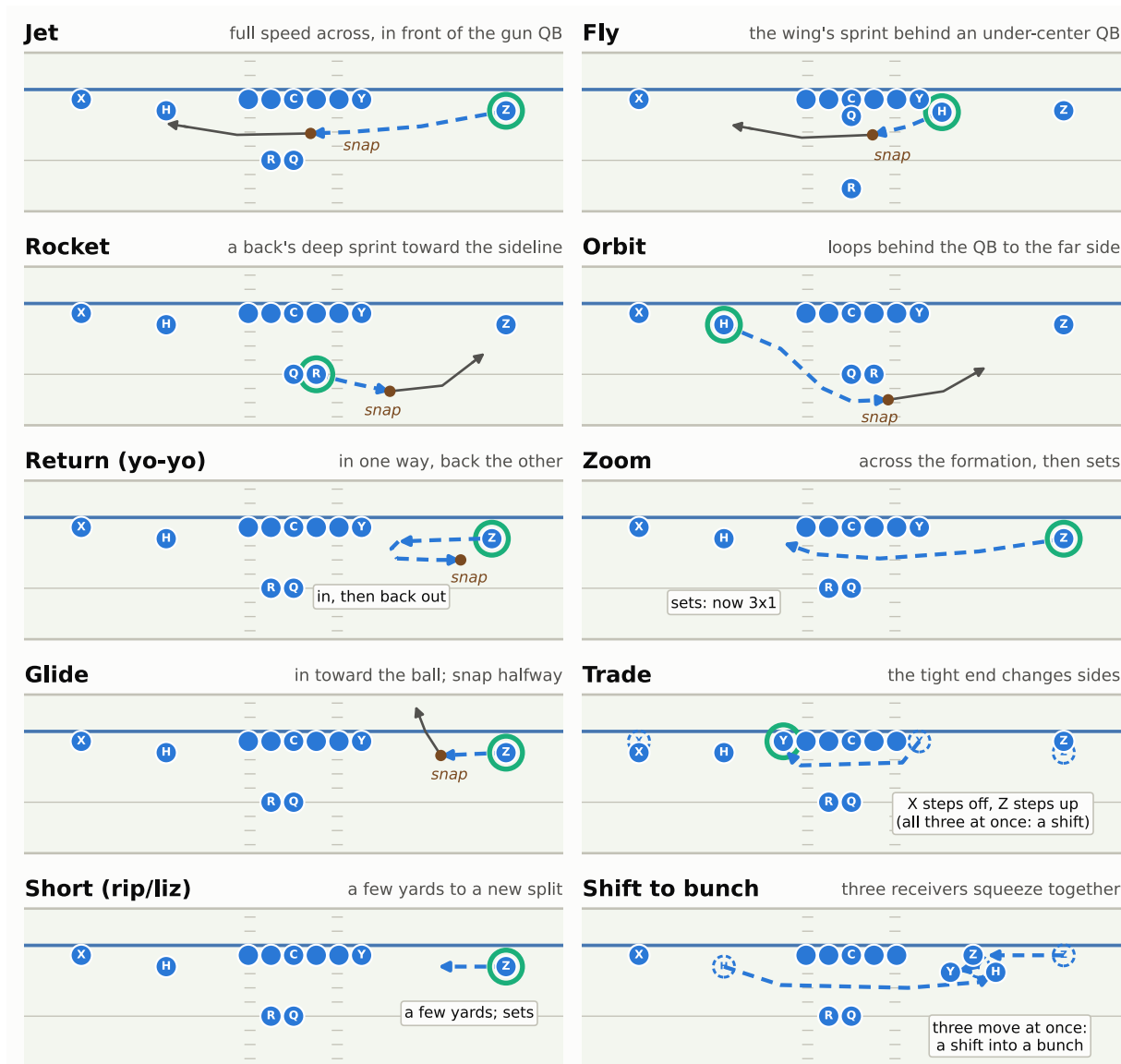


Figure 1: The motion family, offense only, each drawn from a typical alignment. Dashed blue lines are pre-snap movement; thin grey lines show what the player does after the snap; a brown dot marks where he is when the ball is snapped (motions with no dot stop and set before the snap). Jet, fly, rocket, orbit, glide and return are usually run as motion at the snap; zoom and short motion usually set first. The trade and the shift to bunch move several players at once, so they are shifts: dashed outlines show where those players started. Names vary from playbook to playbook; the shapes are what matter.

Common misconception

“Jet motion” sounds like a technical term with one meaning. It isn’t. Motion names are **playbook dialect**: each staff names its motions in its own language, often after the letter of the player who moves (“zip” and “zoom” for the Z receiver, “yo-yo” for Y), and the same word can mean different things in two buildings. Some teams use *jet* and *fly* for the same motion; some use *orbit* only for a running back. The names below are the common ones in NFL film breakdowns and coaching clinics, and the glossary records them.² Learn the **shapes**, and you’ll recognise the motion whatever the broadcast calls it.

Motions that cross the formation. These are the big, fast ones, almost always timed so the ball is snapped while the player is still moving.

²Common definitions of jet, return, orbit and glide motion as coaches teach them: Ryan Swingle, “Install Motion Into Your Offense (Coaching Guide),” Viqtory Sports (updated Jan. 27, 2026): <https://www.victorysports.com/pre-snap-movement-in-the-offense/>. The guide notes that “fly” is used for jet motion in some terminology, especially the Wing-T.

- **Jet motion:** a receiver sprints across the formation, flat along the line, timed so that the ball is snapped just *before* he reaches the quarterback. In the shotgun he passes between the quarterback and the line, close enough to take a handoff or a little forward flip (the jet sweep, a run to the edge by the motion man himself). Most of the time he doesn't get the ball: the jet is a *threat*, and the defense has to respect it either way.
- **Fly motion:** the older name for the same idea from the Wing-T, a run-heavy offense of the 1950s onward. A wingback (a back lined up just outside and behind the tight end) sprints across behind an under-center quarterback, who turns and hands him the ball on the "fly sweep." Many staffs say *jet* and *fly* interchangeably. Where they distinguish them, *fly* is usually the under-center version, with the runner passing behind the quarterback.
- **Orbit motion:** a receiver comes in toward the ball and swings *behind* the quarterback (and behind the back), like a moon around a planet, and is still moving toward the far side at the snap. Going behind the quarterback means he can take a pitch or a shovel pass, or turn into a receiver out of the backfield, and he is harder for the defense to track because he disappears behind the offensive line for a moment.
- **Zoom motion:** a receiver goes all the way across the formation and then *sets* on the other side. It isn't about speed; it is about the new formation. A 2x2 becomes a 3x1, and the defense has to rebalance (the job in Job 2, below).

Motions out of the backfield.

- **Rocket motion** (also **rock motion**): a back sprints from his spot in the backfield straight out toward the sideline, staying deep, so he is already running wide when the ball is snapped. In option offenses such as the flexbone the same name is used for a slotback (a back set just behind the line, outside the tackle) sprinting on a deep path behind the quarterback.³ The play it sets up is the **rocket toss**, a pitch to the back at the edge, and the rocket motion is a fine way to get a back into a pass route with a head start, too.

Motions that come back, or come in. These are subtler, and they matter most in the passing game.

- **Return motion** (also **yo-yo motion** or **yo-yo**): a receiver motions one way, then reverses back toward where he started, usually right at the snap. Its targets are the defenders who follow motion: whoever chased the first move is now running the wrong way.
- **Glide motion:** a wide receiver motions in toward the ball at a controlled pace, and the snap comes when he is about halfway to the formation. He starts the play already moving inside, which helps a crossing route, a quick break back outside, or a crack block (a receiver blocking back inside on a linebacker or safety, taught with the perimeter runs in Perimeter Runs and Misdirection).
- **Short motion** (also **rip/liz motion**, after the "right" and "left" tags some playbooks use): a few yards and no more, usually a receiver tightening his split, or tucking in beside or behind a teammate to form a stack, and usually set again before the snap. Short motion looks like nothing, which is part of the point: a receiver who starts closer to the ball has more room to the outside, and a defender who lined up against his old spot is a step out of place.

Moves that change the whole formation. These are usually *shifts*, because more than one player moves.

- A **trade** (a **TE trade**): the tight end changes sides. Because he leaves the line, someone else has to keep the offense legal (seven on the line, with only the end men eligible): usually a receiver on the new side steps *off* the line and one on the old side steps *on*, all at once. The strength flips, and the defense has to flip its front and its coverage calls with it. (You saw a trade, and the defense flipping in answer, in The Pre-Snap Phase.)
- The **shift to bunch**: three receivers squeeze into a tight triangle at the last moment. Against man coverage a bunch is a traffic jam that defenders have to fight through (the crossing routes it makes possible are the rub routes of the passing chapters), and creating it *late* means the defense can't line up in its planned answer to bunch until it's almost too late to communicate it.

And the timing word: motion at the snap. **Motion at the snap** isn't a path. It means the player is *still moving* when the ball is snapped. Jet, fly, rocket, orbit, glide and return motion are almost always run this way; zoom and short motion usually aren't. The distinction matters so much in the modern game that it gets its own section later in the chapter.

i Madden vs. real life

In *Madden*, you can send almost any player in motion from a menu before any play, in any direction you like. Real NFL motion is almost never improvised. It is written into the play call ("Gun Trips Right, Z Jet, 26 Zone"), rehearsed dozens of times in practice so that the snap arrives when the motion man reaches an exact spot, and paired with the play it is meant to help. The quarterback's job isn't to *choose* the motion. It's to *read what the defense does* with it, and to snap the ball at the right instant.

³In flexbone and Wing-T offenses the "rocket toss" sends a slotback in fast motion on a deep path behind the quarterback for a pitch; shotgun offenses use "rocket" (and "laser" for the other direction) for a back's fast lateral motion out of the backfield. Swingle, Victory Sports (above).

Job 1: asking man or zone

Start with the question in the opening scene. When a receiver goes in motion, the defense has three basic ways to deal with him, and each one looks different.

Travel: somebody goes with him

In man coverage each defender has a receiver, so the simplest answer to motion is for that defender to go with him. A defender who follows a man in motion across the formation is said to **travel** with him.

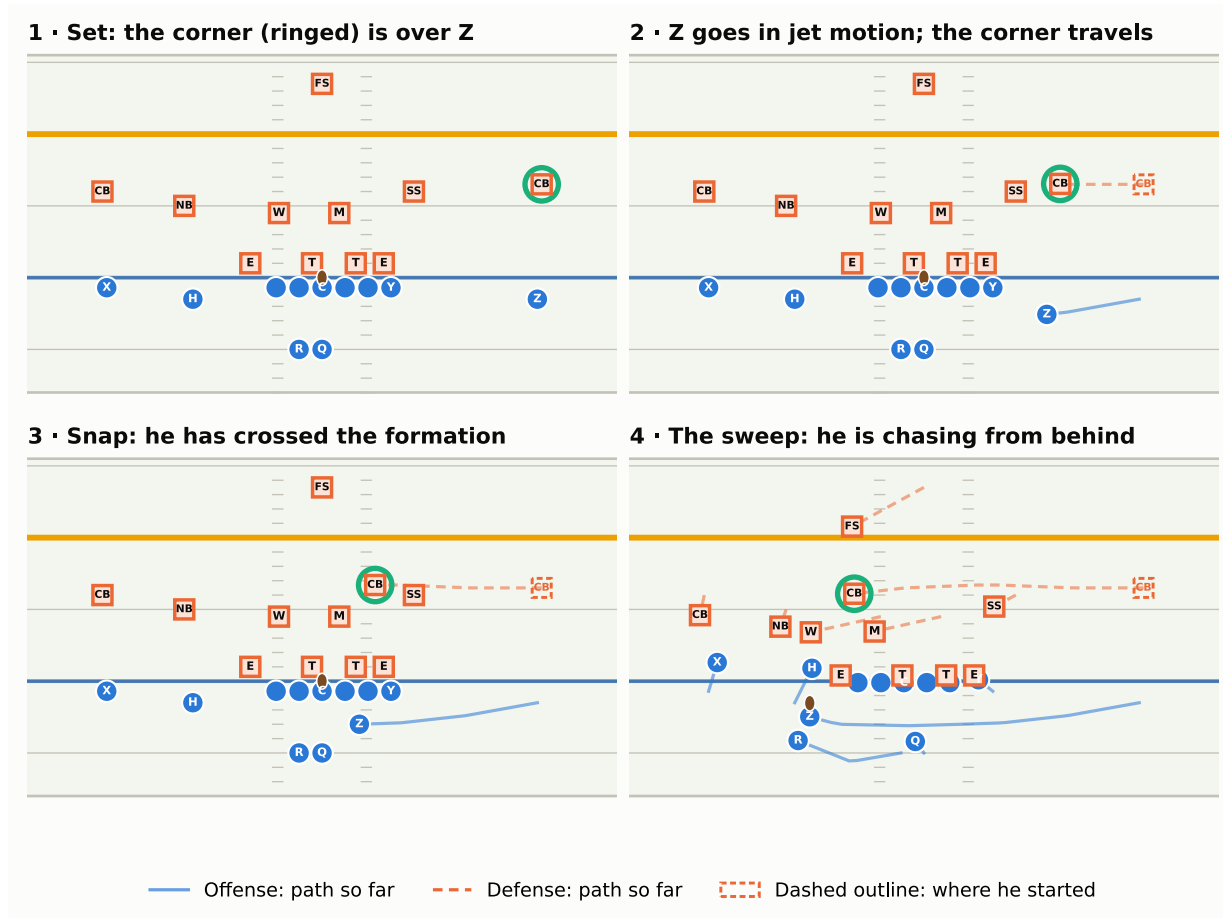


Figure 2: Jet motion against man coverage (simulated, in Big Data Bowl tracking coordinates). The defense is in Cover 1: each defender has a receiver, and one free safety (FS) stays deep in the middle. When Z sprints across in jet motion, his cornerback (ringed) travels with him, step for step: that is the man coverage giving itself away. At the snap the corner is on the wrong side of the formation, inside and behind a receiver who is already at full speed. The ball is flipped to Z on a jet sweep and the corner can only chase. Faint lines are paths so far; the dashed outline is where the corner started.

To the quarterback, a corner running across the formation is about the loudest signal a defense can send. A zone defender has no reason to leave his area to follow one receiver across the field, so a traveller almost always means **man coverage**.

Why travel at all, if it's so obvious? Because it keeps the defense simple. The matchups the coaches planned all week (our best corner on their best receiver) survive any motion, and nobody has to make a new call in the last two seconds. That's a real advantage against a team that motions on most snaps.

The price is in the fourth panel. The traveller is now **inside and behind** a receiver who is already running at full speed, and he started the play moving *sideways*. Anything that goes toward the side the motion is heading (a jet sweep, a flat route, a wheel up the sideline) is a race the defender begins already losing. And when the corner crosses the formation he ends up somewhere he wasn't built to play, near the box, where a run might come straight at him. That's why many man defenses don't travel on every motion: on a full-speed jet across the formation, they trade assignments instead (more on that in a moment).

Bump and spin: the zone answer

A zone defense doesn't care who the receiver is; it cares where he is. So when a receiver crosses the formation, nobody follows him. The defenders **adjust their spacing**, each moving over one spot toward the motion, and the safeties may rotate so that the side the motion is heading to gets an extra man.

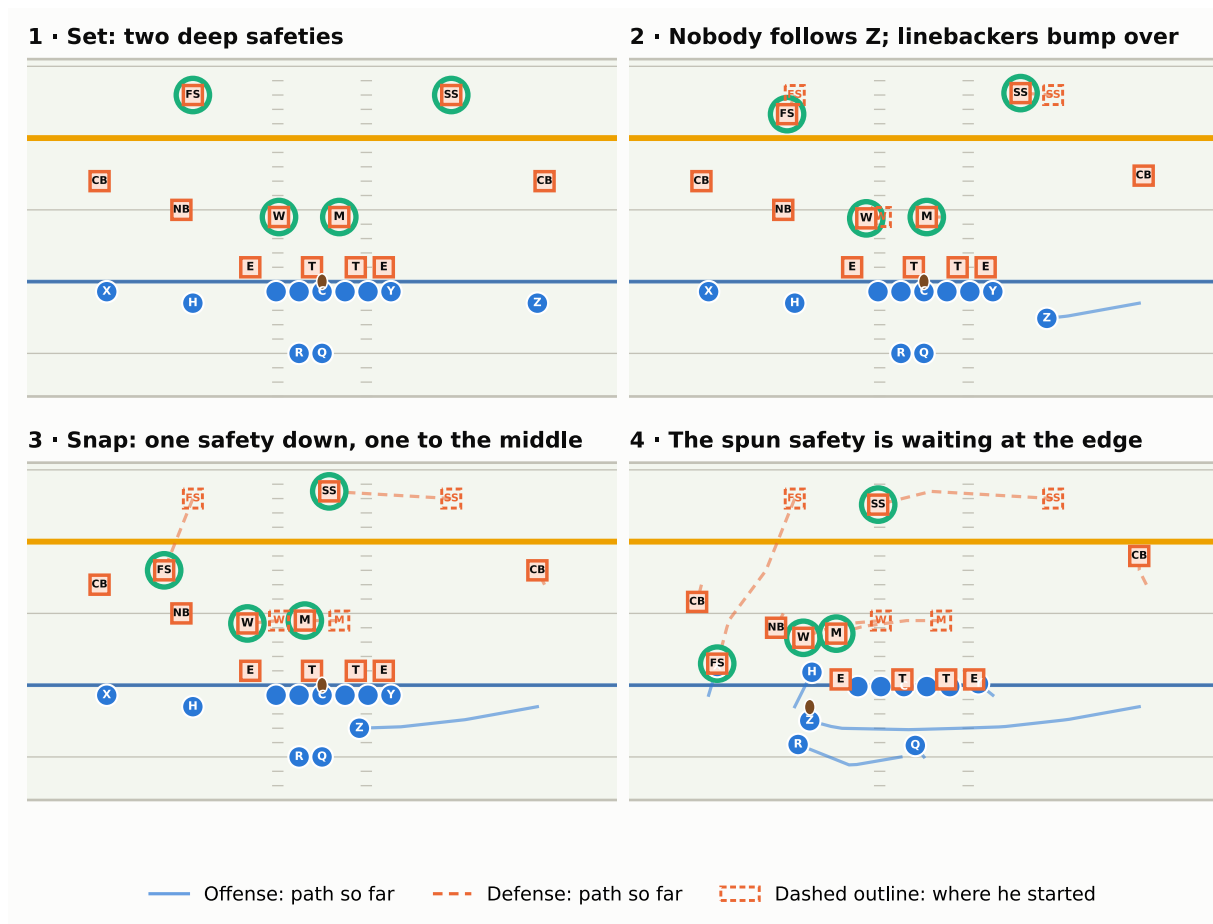


Figure 3: The same jet motion and jet sweep against a zone defense (simulated, in Big Data Bowl tracking coordinates). The defense starts two-high: one deep safety on each side. When Z goes in motion, nobody follows him. The two linebackers bump a step toward the motion, and the safeties spin: the one on the side Z is heading to (FS) comes down toward the line, and the other (SS) rotates to the deep middle, so the shell turns into one-high. The right corner stays home over empty grass. At the snap the defense has an extra player on the side of the sweep, and the safety who spun down is waiting at the edge. Ringed players are the ones who moved; dashed outlines are where they started.

Two things happened, and both have names.

- **The bump.** The linebackers each slid a step toward the motion. A **bump** is defenders shuffling one spot toward the motion instead of following it, each taking over a new area or a new receiver. Nobody crossed the formation; everyone just *re-spaced*. Coaches also call it a slide or a shuffle.
- **The spin.** The two safeties rotated. A **spin** is one safety coming down toward the side the motion is heading to while the other moves to the deep middle, so two deep safeties become one. (The rotation itself, and every variety of it, is taught in Disguise, Rotation, and Split-Field Coverage.)

Why answer motion this way? Because it keeps the *numbers* right without sending anyone on a long run. After the motion there are more receivers on the left, so the defense moves more men to the left. The safety who spins down becomes an extra defender on the motion side, the man who stops the jet sweep in the fourth panel. Every receiver is still inside somebody's area, and no defender ended up on the wrong side of the formation.

The price: the defense has shown its hand. Nobody travelled, so it's probably zone. The safeties spun, so it's probably one-high now, with three deep defenders (the two corners and the middle safety) and four or five underneath, the shape of Cover 3. The side the motion *left* is thinner than it was, with a corner standing alone over empty grass. And the spin took time: a safety who

must cover the deep middle had to run eight or nine yards before the snap, which is exactly why offenses love to snap the ball while he is still running.

The man/zone indicator, and the running back version

Put the two pictures together and you have the **man/zone indicator**: reading the defense's reaction to motion as a clue to its coverage.

Table 1: Reading the defense's answer to motion. Each row is a hint, not a guarantee; the next section explains why.

What the defense does with the motion	What it suggests
One defender runs with the motion man (travel)	Man coverage
Defenders shuffle over without following (bump)	Zone coverage
A safety comes down toward the motion and the other goes to the middle (spin)	A rotation to one-high (often Cover 3, or man with one deep safety if someone also travelled)
Safeties don't move at all	The shell you see may be the shell you get, or the rotation is coming at the snap

The cleanest version of the test doesn't use a receiver at all. Motion the **running back out of the backfield** to a wide spot, leaving the quarterback alone (an empty formation). In man coverage, someone is assigned to that back, usually a linebacker, and that linebacker has to walk out of the box with him. In zone, the defender already standing out there (a nickel back or an outside linebacker) simply takes him, and the linebackers stay home. A linebacker walking out to the numbers with a running back is one of the most reliable pre-snap tells in football. It also tells the quarterback that the middle of the field has one fewer defender in it, which is why quarterbacks who can run love to see it.

! Why it exists

The man/zone indicator exists because **coverage is the defense's best-kept secret, and motion is the cheapest way to make it leak**. A defense can hide its coverage while everyone stands still. It can't avoid *reacting* to a player who moves, and each way of reacting has a different shape. For the offense the information is worth a lot: man coverage invites crossing routes, rubs and quarterback runs; zone invites throws into the gaps between defenders. With a kill call or a built-in check, the quarterback can swap the play for one that suits the answer he just got.

When the indicator lies

Defensive coordinators know exactly what you just learned, and they spend their careers making it unreliable. Three counters are worth knowing now; the full treatment is in Disguise, Rotation, and Split-Field Coverage.

Man defenses that bump. A man defense doesn't have to travel. Many play man with a rule that says, in effect, "if he crosses the formation, *hand him off*": the defender over the motion man stays put and takes whoever ends up in front of him, and a defender on the other side picks up the motion man. The coverage is still man, but nobody ran across the field, so it looks like a zone bump. Defenses often decide this by the *kind* of motion: they might travel on a short motion (cheap to follow) and exchange on a full-speed jet (expensive to chase).

Zone defenses that follow. Modern zones are pattern-matching coverages: each defender starts with an area but plays the receiver who comes into it like man coverage. Under some of those rules a defender follows the motion man a few steps before passing him on. The quarterback sees a defender move with the motion and thinks *man*; the coverage is zone.

Fake spins. A safety spins down with the motion, the quarterback concludes "one-high," and at the snap the safety backs out again into a two-high shell. Defenses that disguise heavily rehearse every rotation from several starting pictures for exactly this reason.

That last counter is why offenses have pushed motion closer and closer to the snap. If the ball is snapped while the motion man is still running, the defense has to react *once*, at full speed, with no time for a second move. A defender who travels at the snap really is chasing; a safety who spins at the snap really is rotating. Motion at the snap doesn't make the defense tell the truth, but it leaves much less time to lie.

Job 2: changing the strength and the numbers

The second job of motion is simpler and older: change the formation after the defense has lined up against it.

Defenses align to the offense's **strength** (usually the tight end's side, or the side with more receivers), and they assign their pass defenders by counting receivers on each side. When a player crosses the formation, both of those change, and somebody on the defense has to move.

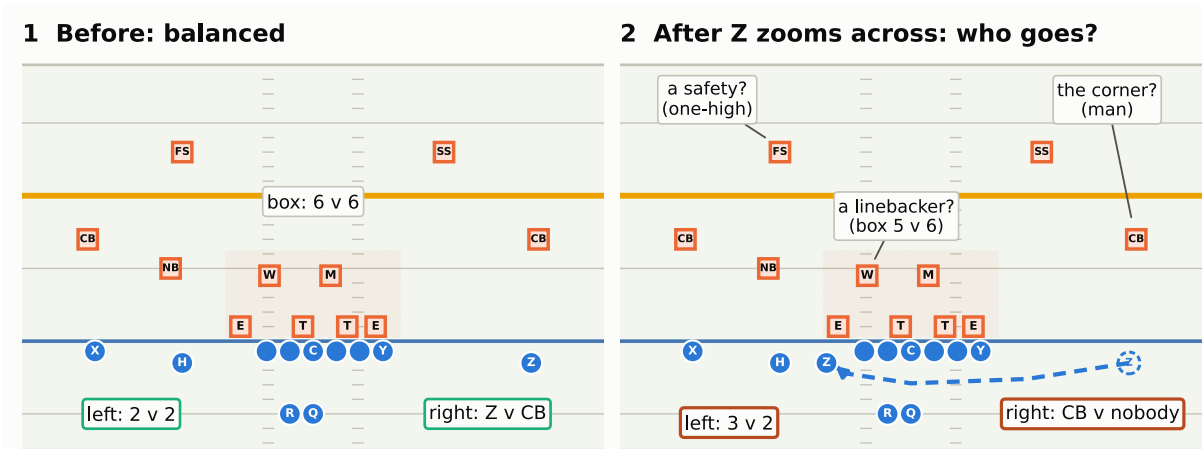


Figure 4: Motion forces the defense to rebalance. Left: a 2x1 formation with the tight end (Y) on the right, against a two-high nickel defense. Counting receivers outside the box, the left side has two receivers (X and H) and two defenders over them (CB and NB); the box is six blockers (five linemen and Y) against six defenders. Right: Z zooms across and sets on the left, making it 3x1. If the defense doesn't move, it is three receivers against two defenders on the left, and a corner on the right guarding nobody. Someone has to go left: a linebacker (and the box drops to five against six, good for the run), a safety (and the shell turns one-high), or the corner himself (man coverage, and the travel gives it away). Whichever moves, the spot he left is weaker.

This is the arithmetic every motion forces. **The defense has eleven players, and each one it moves toward the motion comes from somewhere.** If a linebacker walks out, the box gets lighter, and the run game gets the numbers advantage that The Pre-Snap Phase taught you to count. If a safety rotates over, the deep coverage changes and the lone receiver on the other side may be in a one-on-one with no help over the top. If the corner follows, the coverage is man and the quarterback knows it. Offensive coordinators design motions to make the defense choose among those three, then call plays that attack whichever spot the defense gave up.

The **trade** does the same thing to the defensive *front*. Defensive linemen and linebackers set their alignment to the tight end: the bigger end lines up across from him, the strong-side linebacker walks out to his side, and the strong safety cheats toward him. When the tight end trades sides, all of that has to flip in the two or three seconds before the snap, which is why offenses that like to run at the strength love to trade at the last moment. A defense that flips late and loses a gap in the confusion is a run waiting to happen; a defense that refuses to flip has its strength on the wrong side.

The **shift to bunch** attacks the coverage the same way. Most defenses have a pre-planned answer to a bunch, because man coverage against three receivers in a tight triangle invites collisions and picks. If the bunch appears with two seconds left, that answer has to be called, heard and applied by four or five defenders at once.

Job 3: leverage and the running start

The third and fourth jobs are about the moment the ball is snapped, and they are really one idea: **a player who is moving at the snap begins the play somewhere the defense didn't plan for, at a speed the defense doesn't have.**

The running start

How big is a running start? You can estimate it with simple physics. A football player accelerates roughly like a sprinter: hard at first, then levelling off near his top speed. The chart below compares two receivers with the same top speed, one starting from a standstill and one already running at jet-motion pace when the ball is snapped.

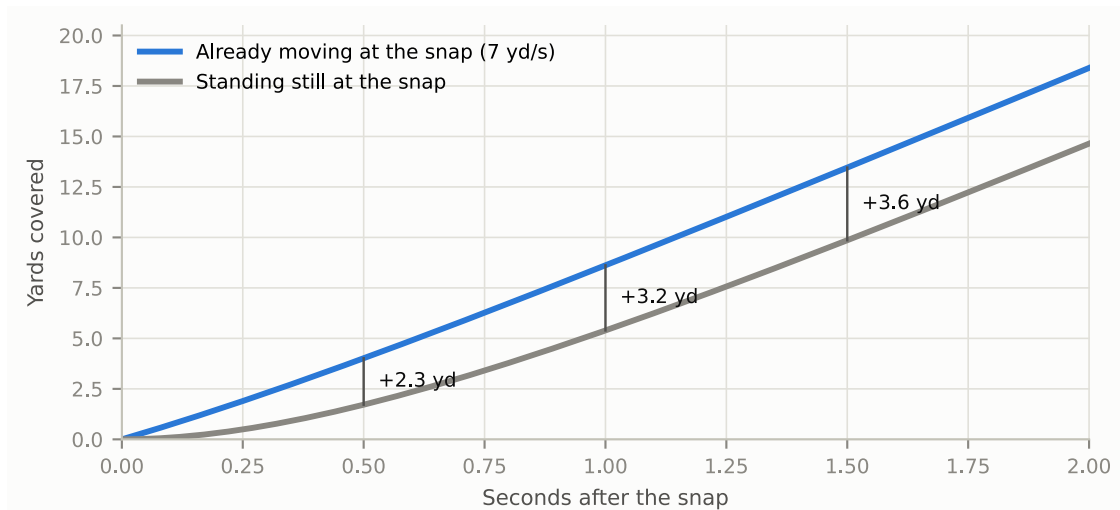


Figure 5: The running start, in a simple model. Each receiver accelerates toward a top speed of 10 yards per second (about 20 mph) with a time constant of 0.55 s, which gives a 40-yard dash of about 4.5 seconds from a standstill. The standing receiver starts at 0; the one in jet motion is already moving at 7 yards per second when the ball is snapped. After one second the moving receiver has covered about 3 more yards than the standing one, and he keeps nearly all of that lead. A defender who reacts to the snap starts from zero as well, so motion at the snap turns a race into a head start. This is an illustrative model, not tracking data; the real numbers vary by player and by how fast the motion is.

Three yards in a second is enormous. It is the difference between a jet sweep that reaches the edge before the defensive end can widen and one that is tackled for no gain; between a receiver who is covered and one who is open. A defender can match a receiver's speed. He can't match a head start, because he isn't allowed to have one: he must react to the snap like everybody else.

The same physics explains why offenses with very fast receivers motion so much. A head start is worth more to a receiver with a high top speed, because he gains more ground during the acceleration that the defender is still going through. It's also why defenses hate pressing a receiver who is in motion: press coverage (a defender lined up right on the receiver to jam him at the line) depends on getting hands on a man who is standing still. You can't jam somebody who is already running past you.

Leverage: start where he didn't plan for

Every defender aligns with leverage, a deliberate position inside or outside his receiver that tells him which way to force him and where his help is. That alignment is planned against a receiver who is standing still. Motion changes it in a way the defender can't fully fix:

- A defender who **travels** with motion is, by definition, *behind* the receiver in the direction he is moving. A receiver moving inside at the snap (glide motion) is already inside his defender, which helps routes that break inside. A receiver moving outside at the snap (orbit, rocket) is already outside his.
- A defender who **bumps** is standing in a spot chosen for the *old* formation, a step or two from where he would have lined up if the offense had started in the new one.

The most famous way to exploit the first of these is the return motion.

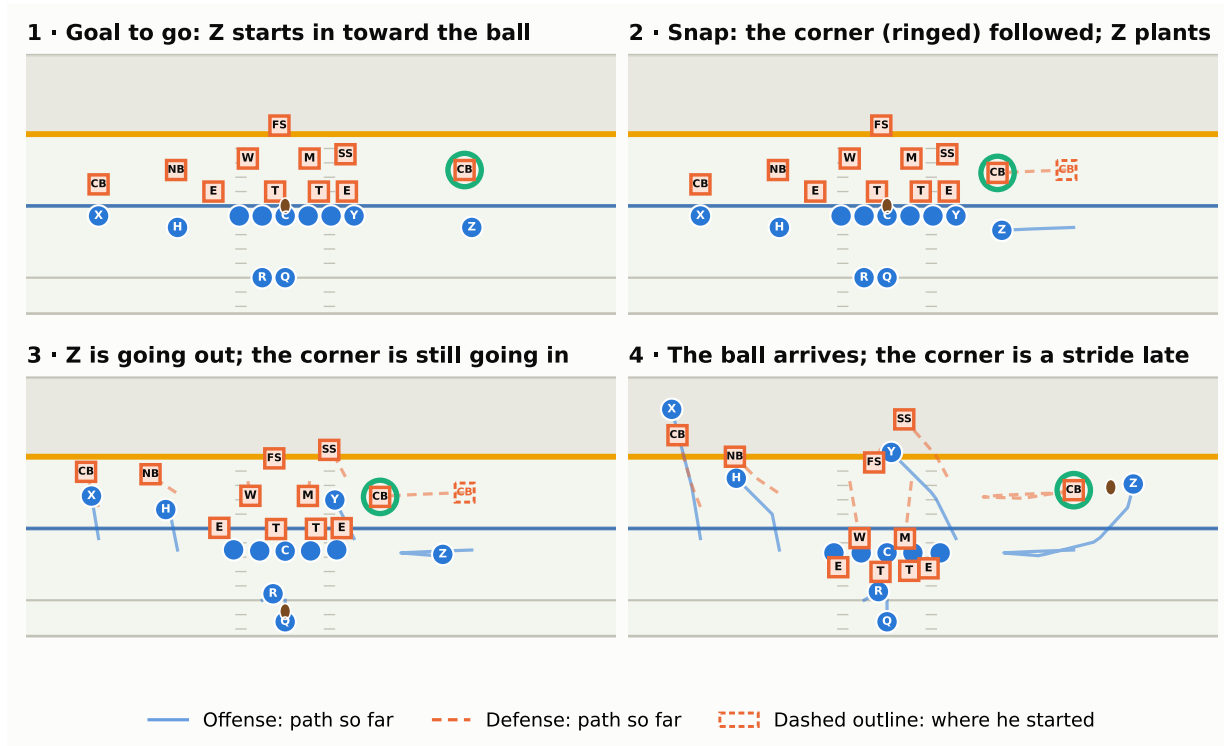


Figure 6: Return motion against man coverage at the goal line (simulated, in Big Data Bowl tracking coordinates). The defense is in Cover 0: man coverage with nobody deep, and six rushers. Z motions in toward the ball and his cornerback (ringed) travels with him, still moving inward when the ball is snapped. Z plants and runs back out toward the pylon. The corner has to stop, turn around and restart, and by the time the quick throw arrives he is a stride behind. This is the idea behind the Chiefs’ ‘Corn Dog’ touchdowns in Super Bowl LVII (see the film room); the diagram is illustrative, not those plays.

Look at the defender’s problem in the third panel. A man defender’s whole technique is to mirror the receiver, and here the receiver has reversed direction at the one instant the defender can’t anticipate. Every step the corner took following the motion is now a step in the wrong direction. Against zone the same motion does little (the zone defenders bump and then sit in their areas), which is why return motion is called most often when the offense expects man coverage: near the goal line, on third-and-short, against blitzes.

The jet as a fake

Most jet motions end with the ball going to somebody else. That’s not wasted motion. A receiver sprinting across the formation at the snap is a threat to run a sweep, so the defenders responsible for the edge on that side must take a step toward him, and the linebackers’ eyes go with him. A run the other way, or a play-action pass behind the flowing linebackers, now has one fewer defender in its way, or one who is a step late. Offenses that run a lot of jet motion build their plays in pairs: the jet sweep itself, and the plays that take advantage of the defense *respecting* the jet sweep. That pairing, and why the sweep has to be run often enough to stay believable, is the subject of Perimeter Runs and Misdirection.

Job 4: stressing the defense’s communication

The fifth job doesn’t show up on a diagram, but defensive coaches talk about it constantly. Every change in the formation obliges the defense to **re-communicate**: the linebacker who sets the front re-declares the strength, the safeties re-call the coverage check (many coverages change when the formation goes from 2x2 to 3x1), the corners confirm who has whom, and in a man defense the players involved have to agree in one word whether they’re travelling or exchanging.

All of that happens with the offense deciding when the ball moves. If the motion starts with ten seconds on the play clock, the defense has time. If it starts with three, and the ball is snapped as the motion man passes the quarterback, the defense has one chance to get every call right, often in a loud stadium. The typical failure isn’t dramatic: two defenders take the same receiver and nobody takes another, or a safety rotates the wrong way. But one blown assignment is usually all an NFL offense needs for a big play.

The stress runs both ways. The offense, too, has to execute the motion perfectly: the motion man has to start on the right word, run at the right pace and be at the right spot when the ball is snapped, and the center has to snap it at precisely that moment. A

motion that arrives early leaves the receiver standing and waiting; one that arrives late leaves the quarterback holding the snap while the defense sorts itself out.

Motion at the snap

Everything above points to one conclusion: **the later the motion, the more it's worth**. A motion that ends two seconds before the snap gives the defense two seconds to adjust, disguise and re-communicate. A motion that is still going when the ball is snapped gives it none. Through the late 2010s and 2020s, NFL offenses learned that lesson fast.

ESPN Analytics, which charts specifically whether a player is moving at the snap, found that the average team did it on about **4%** of its plays in 2017 and about **22%** in 2023.⁴ The 2023 Miami Dolphins, coordinated by head coach Mike McDaniel, were the extreme: early that season ESPN had them with a player in motion at the snap on **59%** of plays, by far the most of any team since ESPN began tracking in 2017.⁵ By 2025 other offenses had caught up: according to Next Gen Stats, the Atlanta Falcons under offensive coordinator Zac Robinson led the league in 2025 with motion at the snap on **60.8%** of their plays.⁶

Why is motion at the snap so valuable? Everything in this chapter so far, at once:

- **A running start**, at full value, because the motion man is at speed when the ball moves.
- **No time to disguise**. The defense's reaction to the motion happens at the same instant as its reaction to the snap, so a fake spin or a late travel has nowhere to hide.
- **Leverage the defender can't fix**, because he can't re-align after the snap.
- **The communication stress is at its peak**, because the adjustment has to happen in real time.

And what does it cost? The motion man can't be a pass blocker, and his route has to start from wherever the motion takes him. The timing has to be perfect on every snap. And it hands the defense one piece of information that defensive coordinators have learned to use: **if the snap comes when the motion man reaches the quarterback, the defense knows when the snap is coming**. A defense that times its own movement (a blitz creeping up, a safety rotating) to the motion man's arrival can move at the last instant without risking an offside penalty. Writing in January 2024, Ben Solak of *The Ringer* pointed out that the two defenses then meeting in the AFC Championship Game, Steve Spagnuolo's Chiefs and Mike Macdonald's Ravens, ranked first (Baltimore) and sixth (Kansas City) in EPA per play allowed on snaps with motion, according to Next Gen Stats, and that both made a habit of changing the whole picture as the motion triggered the snap.⁷ The arms race moves to the last half-second.

The "cheat motion" and the 2024 rule language

Motion at the snap reached the edge of the rulebook in 2023. The Dolphins (and soon the 49ers and Rams) had receivers who started on the line of scrimmage take a short step or two *backward and in*, so that they were already in motion, and moving toward the line again, as the ball was snapped: a sprinter's rolling start. Kyle Shanahan called it a "little cheat motion" and admitted the 49ers had copied it "in Week 2 after we saw it."⁸ Tyreek Hill used it all season without drawing a single illegal-motion flag.

Was it legal? The rule said a player may not be moving *toward* the line at the snap, and the officials judged that most of these motions were sideways or backward at the moment the ball moved. For 2024 the league added language aimed squarely at the trick. An eligible receiver who is **on the line** and moves backward or changes his stance now has to come to a **complete stop** before the snap, or it's a false start; a player in the backfield who changes his stance doesn't have to stop, as long as his movement isn't abrupt (a false start) or forward (illegal motion).⁹ The running start survived; the rolling start from the line didn't.

⁴ESPN Analytics (Seth Walder), as reported by Ben Solak, *The Ringer* (Jan. 25, 2024): an average NFL team snapped the ball with a player in motion on 4% of plays in 2017 and 22% in 2023: <https://www.theringer.com/nfl/2024/1/25/24050457/kansas-city-chiefs-baltimore-ravens-afc-championship-steve-spagnuolo-mike-macdonald-defense-motion>. An ESPN figure of about 25% for 2024 is reported second-hand.

⁵Seth Walder, ESPN, reported via 4for4 (Sept. 26, 2023): "The Dolphins have had a player in motion at the snap on 59 percent of plays this season," the most since ESPN's tracking began in 2017 (league about 20% at the time): <https://www.4for4.com/news/dolphins-using-motion-more-any-other-team>. FTN figure (79.4% of 2023 regular-season runs and passes with *is_motion*) is a coarse calculation from nflverse.

⁶Brianna Dix, "Rise of motion in NFL offenses," *Buccaneers.com* (Feb. 3, 2026), citing Next Gen Stats: Atlanta under Zac Robinson ranked first in motion at the snap in 2025 (60.8%) and sixth in pre-snap motion (70.1%): <https://www.buccaneers.com/news/rise-of-motion-nfl-offenses-what-it-means-bucs-2026-robinson>

⁷Ben Solak, *The Ringer* (Jan. 25, 2024), on the Chiefs' and Ravens' defenses "changing the picture" as motion triggered the snap, and their Next Gen Stats ranks (Ravens first, Chiefs sixth) in EPA per play on snaps with motion; URL in the motion-at-the-snap note above.

⁸John Breech, "NFL shuts down rulebook loophole that Dolphins, 49ers and Rams were exploiting during 2023 season," CBS Sports (Aug. 1, 2024), including Shanahan's "a little cheat motion" and "We did it in Week 2 after we saw it": <https://www.cbssports.com/nfl/news/nfl-shuts-down-rulebook-loophole-that-dolphins-49ers-and-rams-were-exploiting-during-2023-season>; Alain Poupart, *Sports Illustrated* (All Dolphins), on Hill drawing no illegal-motion flags in 2023: <https://www.si.com/nfl/dolphins/tyreek-and-the-commotion-about-the-cheat-motion-01j47qrysqj>

⁹NFL rule language added for 2024 (false start, Rule 7, Section 4, Article 2), as quoted by CBS Sports (Breech, above): "If an eligible receiver who is on the line of scrimmage moves backward or changes his stance (two point to three point or conversely) and does not come to a complete stop before the snap, it is a false start. Any eligible backfield player who changes his stance does not have to come to a complete stop prior to the snap, as long as his actions are not abrupt (false start) or forward (illegal motion)."

The rulebook: NFL, college and Canada

The NFL's motion rule was covered in The Pre-Snap Phase: **one** player in motion at the snap, he must be a **backfield** player, and he must be moving **parallel to or away from** the line. Two details from this chapter are worth adding.

First, a player who is *on the line* can't go in motion at all without first getting off it, and that has real consequences. In Super Bowl LVII, when the Chiefs wanted to run their return-motion play a second time, Patrick Mahomes signalled for Kadarius Toney to go in motion, but Toney was lined up on the line of scrimmage and couldn't. Toney pointed across the field at Skyy Moore, who was in the backfield, and Moore went instead; he scored the touchdown.¹⁰

Second, the "toward the line" rule is the reason motion is *lateral*. The rule's purpose is the same as the one-second set's: **the offense may move before the snap, but it may not get a running start at the line of scrimmage**. A sideways head start is allowed; a forward one isn't.

Table 2: Motion rules in three codes. American college rules match the NFL's in substance; Canadian rules are different in kind.

	NFL	NCAA (college)	CFL (Canada)
Players in motion at the snap	One	One	Any number of backfield players
Direction	Parallel to or away from the line	Not toward the opponent's goal line	Any direction, including toward the line
Who may move	A backfield player	A backfield player	Backfield players (not the quarterback)

College football's rule is essentially the same as the NFL's: one player, not moving toward the opponent's goal line at the snap.¹¹ The big outlier is Canadian football. In the CFL, every backfield player except the quarterback may be in motion at the snap, in any direction, including **forward**.¹² Canadian receivers routinely hit the line of scrimmage at full speed as the ball is snapped (a "waggle"), which is one reason the Canadian game, on a bigger field with twelve players a side, is so pass-heavy. If you've wondered what would happen if the NFL allowed forward motion, the CFL is the experiment: receivers that are nearly impossible to jam, and defenders who line up far off the ball to avoid being run past.

The motion boom, by the numbers

Motion is as old as the T formation (the Bears' man in motion in the 1940 championship game is in The Pre-Snap Phase), and some coaches made pre-snap movement a signature long before it was fashionable. Tom Landry's Cowboys of the 1960s to 1980s famously had their offensive linemen stand up in unison and then drop back into their stances while the backs shifted behind them, the "Landry shift," partly to hide the new backfield alignment behind a wall of standing linemen.¹³ But for most of NFL history motion was a tool for particular plays, not a habit.

That changed in the late 2010s. Kyle Shanahan (49ers head coach from 2017) and Sean McVay (Rams head coach from 2017) built offenses in which motion was part of almost every call, and as their assistants became coordinators and head coaches around the league, the habit spread. By ESPN's count, the 49ers used pre-snap motion on 69.1% of their plays from 2017 through 2021, the most in the league, and on 76.9% in 2021 alone, while the league as a whole rose from 34.9% (2017) to 46.8% (2021).¹⁴ PFF's longer shift-or-motion series shows the same climb: 37.6% of plays in 2014, 43.0% in 2018, 55.4% in 2022 and a record **63.9%** in 2025, when every team except the Giants moved someone before at least half of its snaps.¹⁵

¹⁰"Chiefs lined up in wrong formation on pivotal Super Bowl play and Patrick Mahomes still managed to throw a TD," CBS Sports (Feb. 2023): <https://www.cbssports.com/nfl/news/chiefs-lined-up-in-wrong-formation-on-pivotal-super-bowl-play-and-patrick-mahomes-still-managed-to-throw-a-td>

¹¹NCAA, *Football Rules and Interpretations*, Rule 7-1-2 (shifts and motion): <https://www.ncaapublications.com/c-4-rules.aspx> ; summary of the NFL/NCAA/high-school rule (one player, not moving toward the line) in "Motion (gridiron football)," *Wikipedia*: [https://en.wikipedia.org/wiki/Motion_\(gridiron_football\)](https://en.wikipedia.org/wiki/Motion_(gridiron_football))

¹²CFL rulebook, Rule 4 (scrimmage) on backfield motion: <https://www.cfl.ca/rules-and-regulations/> ; "Motion (gridiron football)," *Wikipedia* (forward motion allowed in the CFL and arena football).

¹³ITP Glossary: Landry Shift," Integrated Scouting Academy: <https://scoutingacademy.com/itp-glossary-landry-shift/> (all linemen but the center rise from a three-point stance to a two-point stance and back down before the snap, while the backs shift).

¹⁴Wagoner and Thiry, ESPN (Jan. 29, 2022), as in the first note of this chapter; their figures come from ESPN Stats & Information. FTN's 2024 figure for the Rams (76.7%) is a course calculation from nflverse.

¹⁵Daire Carragher, "NFL pre-snap motion usage reaches new highs across the NFL," PFF (Feb. 23, 2026): shift/motion rate 37.6% (2014), 38.4% (2016), 43.0% (2018) ... 61.5% (2024), 63.9% (2025); every team but the Giants at 50% or more in 2025: <https://www.pff.com/news/nfl-pre-snap-motion-usage-reaches-new-highs-across-the-nfl>

Why the boom? Three reasons, each from earlier in this chapter. **Defenses started hiding their coverage better**: Next Gen Stats measured two-high (split-safety) coverage on 32.9% of plays in 2018 and 42.0% in 2025, the highest since its tracking began,¹⁶ and two-high shells are built to rotate, so offenses needed a way to make them commit. **The head start became more valuable** as offenses filled with very fast receivers. And **it worked visibly** for the offenses that did it most, so it was copied.

Here is how the 32 teams moved, in FTN's charting.

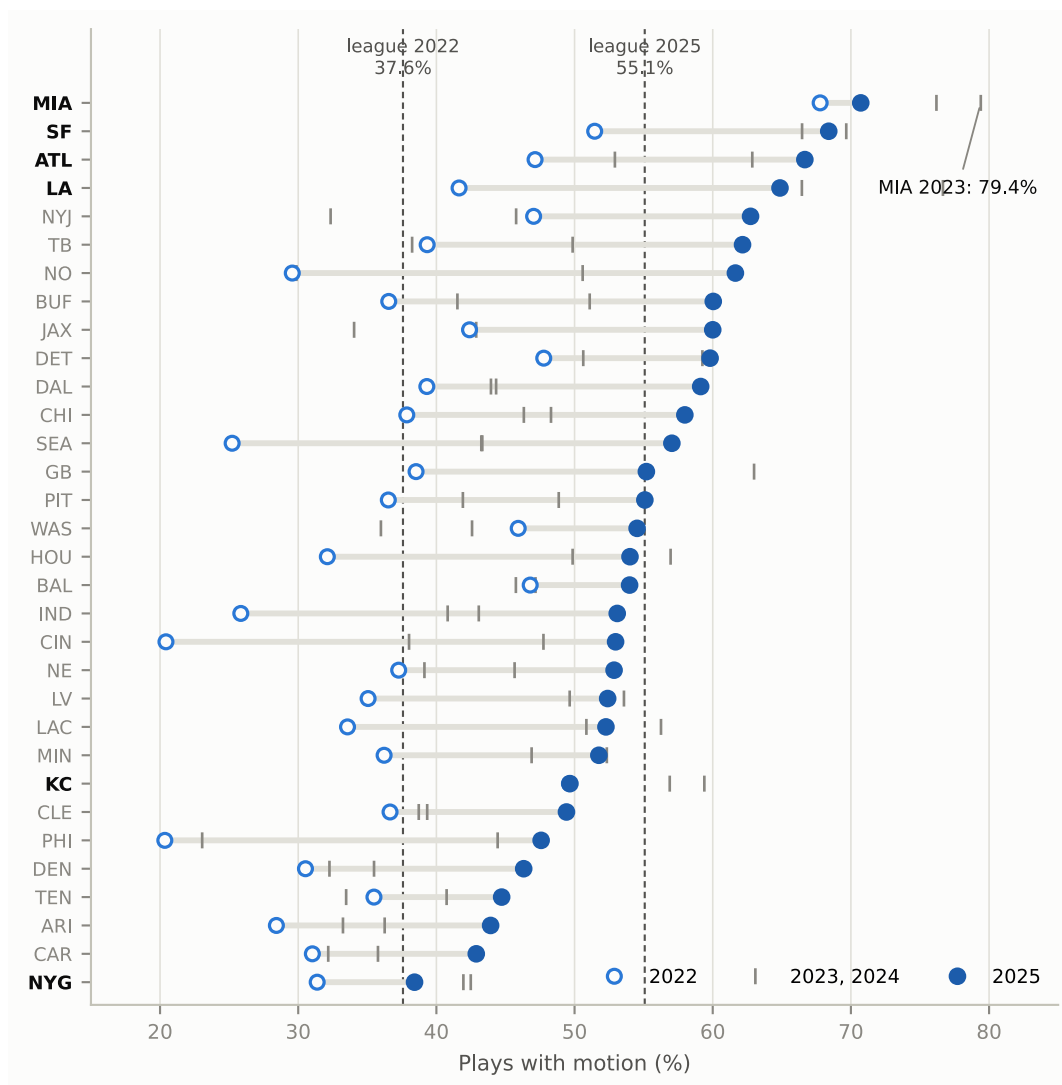


Figure 7: Share of each offense's plays with motion, NFL regular seasons 2022–2025, in FTN's charting (is_motion: motion before or at the snap; runs and passes only). Teams are sorted by 2025. Hollow circles are 2022, small grey ticks 2023 and 2024, filled circles 2025; the dashed lines are the league averages for 2022 and 2025. Almost every offense moved more in 2025 than in 2022. The 2023 Dolphins (79.4%) are the highest team-season in the four years. Only the Giants moved on fewer than 40% of their plays in 2025. Source: nflverse FTN charting and play-by-play; course calculation.

The league average rose from about 38% to 55% in four seasons in FTN's data, and the spread between teams is wide: the Dolphins, 49ers, Rams and Falcons sat near the top for most of the period, while the Giants were the one 2025 offense below 40%. Notice how some teams jumped as soon as they changed play-callers: the Saints went from about 30% to 51% in 2024, Klint Kubiak's first season as their offensive coordinator, and the Eagles from 23% to 44% in Kellen Moore's. Motion is a *coaching* choice more than a personnel one; it follows the play-caller.

Why the motion numbers disagree

You will see wildly different "motion rates" quoted for the same season: 25%, 55%, 64%. They are not contradictions. They are different measurements.

¹⁶Next Gen Stats via NFL.com, "The coverage that turned the Seahawks into Super Bowl winners and is taking the NFL by storm": split-safety coverage 32.9% (2018), 42.0% (2025): <https://www.nfl.com/news/the-coverage-that-turned-the-seahawks-into-super-bowl-winners-and-is-taking-the-nfl-by-storm>

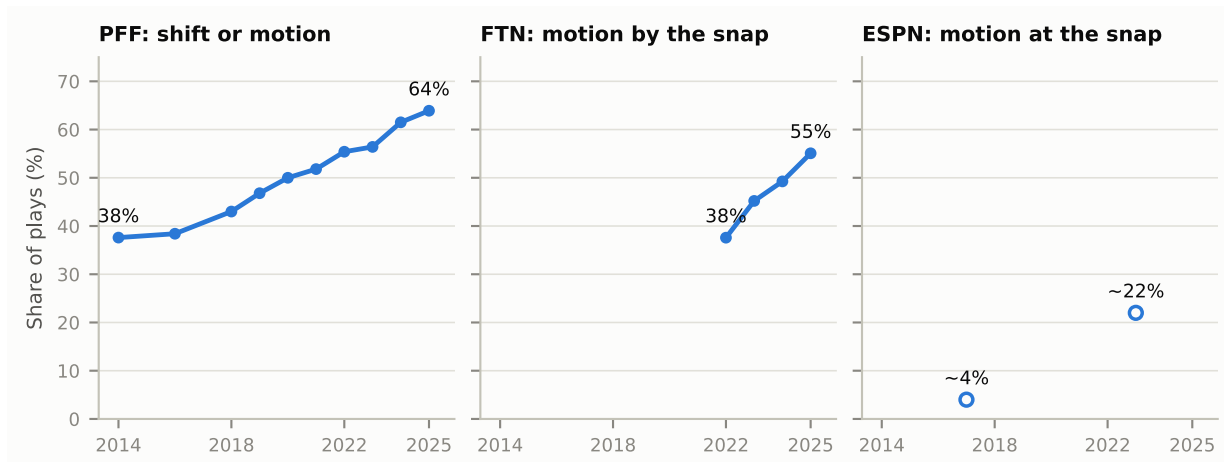


Figure 8: Three published motion series, each on its own panel because they measure different things. PFF counts any pre-snap shift or motion (2014–2025). FTN’s `is_motion` counts plays with motion before or at the snap, charted by FTN (2022–2025, course calculation from nflverse). ESPN Analytics counts a player in motion at the moment of the snap; only two league-average points are published in the sources used here (2017 and 2023), drawn as hollow, unconnected points because they come from press reports, not a data file. All three rise; the levels differ because the definitions do. Never plot them on one axis as if they were one series.

Four things make the numbers differ, and each is worth knowing before you quote one:

1. **What counts as movement.** PFF counts shifts *and* motion. FTN’s `is_motion` flag means “motion occurred on the play before or at the time of the snap.” ESPN’s headline series counts only a player *still moving* at the snap, a much smaller set. Next Gen Stats publishes yet another pair, total pre-snap motion and motion at the snap: Atlanta’s 70.1% and 60.8% in 2025, quoted above.
2. **Who or what does the charting.** PFF and FTN use human charters watching video; Next Gen Stats uses the player-tracking chips. A human charter and a tracking threshold can disagree about whether a receiver’s two-step shuffle “counts.”
3. **Which plays are in the denominator.** All snaps, or only runs and passes? Are kneel-downs, spikes and penalty-only plays included? Playoffs? These shift a rate by a point or two.
4. **Which seasons exist.** FTN’s charting in nflverse starts in 2022, ESPN’s motion-at-the-snap series in 2017, PFF’s runs back further.

The rule this book follows: **every motion number carries its source, and two sources never share an axis.** When you compare teams or seasons, compare them inside one series.

Does motion work?

The honest answer is “yes, but not in the way a simple average suggests.” In FTN’s data, plays with motion and plays without it look surprisingly alike. On first and second downs in 2025, passes with motion averaged about +0.10 EPA per play against +0.06 without, and runs about –0.05 against –0.08; in 2024 the two were nearly identical, and in 2022 passes *without* motion did better.¹⁷ Teams that motion more did have slightly better offenses in each of the four seasons, but the relationship is loose.

Common misconception

“Motion plays gain more, so offenses should motion on every play.” The averages above don’t show that, and even if they did, the comparison wouldn’t prove it. Offenses *choose* when to motion: some save it for their best-designed plays, some motion on almost everything, and defenses adjust to teams that motion a lot. The plays with and without motion aren’t a controlled experiment. Analysts who try to correct for the situation and the play type tend to find a modest benefit; ESPN’s work, as reported by The Ringer in 2024, put motion at the snap at roughly +0.11 EPA per play on runs and +0.08 on passes.¹⁸ Part of motion’s value is also invisible in its own plays: a motion that always meant “run” would be a tell, so offenses use the same motions on different plays to keep the defense from reading them. Measuring that kind of value is part of Tendencies: PROE, Personnel, Motion, and Coverage.

¹⁷Course calculation, nflverse play-by-play joined to FTN charting, regular seasons 2022–2025, runs and passes on first and second down. EPA per play with / without `is_motion`: passes 2022 +0.049 / +0.073, 2023 +0.045 / +0.021, 2024 +0.089 / +0.086, 2025 +0.098 / +0.058; runs 2022 –0.051 / –0.060, 2023 –0.092 / –0.101, 2024 –0.081 / –0.067, 2025 –0.049 / –0.079. Correlation between a team’s motion rate and its EPA per play (all downs): 0.24 (2022), 0.43 (2023), 0.23 (2024), 0.24 (2025).

¹⁸Solak, *The Ringer* (Jan. 25, 2024), citing ESPN Analytics; URL in the motion-at-the-snap note above.

Film room

i Film room: Miami Dolphins 2023, motion at the snap

No offense in the tracking era moved like Mike McDaniel's 2023 Dolphins. In FTN's charting they used motion on 79.4% of their runs and passes, the highest team-season from 2022 to 2025; ESPN had them with a player in motion at the snap on 59% of plays early that season, a record. They scored 70 points against Denver in Week 3, and Tyreek Hill led the NFL with 1,799 receiving yards.¹⁹

What to notice when you watch the film: the motion is short and *timed*. Hill or Jaylen Waddle glides or jogs a few yards, often toward the formation, and the ball is snapped as he gets going. The defense rarely gets a clean look at its own assignment, and Hill starts every route already running. Watch the cornerbacks: against Miami, many played off the line, because pressing a receiver who is in motion is almost impossible. And notice the cost the league imposed: the 2024 language on stops after a backward step from the line was written with this offense in mind. McDaniel was let go by Miami in January 2026 and is the Chargers' offensive coordinator for the 2026 season.

i Film room: Kansas City Chiefs, Super Bowl LVII (February 12, 2023)

Two of the most famous motion plays of the decade came eight minutes apart. In the fourth quarter against the Eagles, Kansas City ran a play its coaches call "Corn Dog": a receiver motions in toward the formation, then turns and runs back toward the sideline at the snap, a return motion built to beat man coverage. Kadarius Toney caught a 5-yard touchdown on it with 12:04 left to give the Chiefs the lead, and Skyy Moore scored from 4 yards on a mirrored version at 9:22. Both times the Eagles were in man coverage, the defender went with the first move, and the receiver was alone when he turned around. Kansas City won 38–35. A year later in Super Bowl LVIII the Chiefs ran a renamed relative of the play for Mecole Hardman's winning touchdown in overtime.²⁰

What to notice: the motion does *two* jobs on one play. First it asks the question (did the defender follow?), and then it attacks the answer (the defender who followed is moving the wrong way). Andy Reid's offense has used motion that way for years, including orbit motion that loops a receiver behind the quarterback to set up pitches, screens and shovel passes back the other way.

i Film room: San Francisco 49ers 2021, Deebo Samuel

In 2021 the 49ers motioned on 76.9% of their plays, the most in the league by ESPN's count. Much of it went through Deebo Samuel, a wide receiver who spent part of the season lining up and motioning into the backfield. Motion let Shanahan move Samuel from receiver to ball carrier and back again before the snap, so the defense never knew which position it was defending. He caught passes for 1,405 yards and ran for 365 more, with eight rushing touchdowns.²¹

What to notice: the motion *creates the matchup*. When Samuel motioned from a receiver spot into the backfield, the defender assigned to him in man coverage had to follow him into the box, where a cornerback or safety is suddenly playing run defense against a 215-pound receiver with a head start. Against zone, the defense now faced a backfield with two ball carriers in it, one of them very fast, and had to respect both.

¹⁹Tyreek Hill, Pro Football Reference: <https://www.pro-football-reference.com/players/H/HillTy00.htm> ; Dolphins 70, Broncos 20 (Sept. 24, 2023): <https://www.pro-football-reference.com/teams/mia/2023.htm>

²⁰"Corn Dog (American football)," *Wikipedia*: [https://en.wikipedia.org/wiki/Corn_Dog_\(American_football\)](https://en.wikipedia.org/wiki/Corn_Dog_(American_football)) (Toney 5-yard TD at 12:04 and Moore 4-yard TD at 9:22 of the fourth quarter; Super Bowl LVIII's renamed version); "The Xs and Os: How the Chiefs won the Super Bowl with a couple of 'Corn Dogs,'" Yahoo Sports: <https://sports.yahoo.com/xs-os-chiefs-won-super-110132971.html>

²¹Deebo Samuel, Pro Football Reference: <https://www.pro-football-reference.com/players/S/SamuDe00.htm>

i Film room: Los Angeles Rams 2017–2025, the same look, different plays

McVay's Rams were not the league's most frequent motion team in his first seasons (ESPN had them eighth from 2017 to 2021), but they made the jet motion famous, and by 2024 FTN charted them with motion on 76.7% of their plays, the highest in the league that season. The Rams' trademark is pairing: a jet motion that comes before an outside zone run one play, a jet sweep the next and a play-action pass the play after that, all from the same look, so the motion tells the defense nothing about the play. What to notice: watch the linebackers when the jet man crosses the quarterback. If they jump toward him, the next jet is likely to be a fake with the run going the other way; if they stay home, the jet man himself gets the ball.

The checklist so far

Part 2 ends here, and with it the first version of the card you will carry through the rest of the book. Each line is a question you can answer before the snap, in the order the answers become visible on a broadcast (the order Personnel introduced): the situation first, then who is on the field, then where they line up, then what the motion draws out of the defense, then the box and the safeties.

Watching checklist • after Part 2		new in Part 2
BEFORE THE SNAP		
1	Situation: down, distance, field zone, clock and score	Part 1
2	Personnel: who is on the field for each side (two digits for the offense; base, nickel or dime)	02-01, 02-05
3	Formation and strength: under center or gun, 2x2 or 3x1, where is the tight end?	02-02 to 02-04
4	Motion: does a defender travel (man), do they bump (zone), does a safety spin (one-high)? Is the ball snapped while he's still moving?	02-06
5	Box count: blockers against defenders near the line (light box: run; loaded box: pass)	02-05
6	Shell: one-high or two-high? Did the motion change it?	02-05, 02-06
AFTER THE PLAY		
7	Log it: your run/pass and man/zone calls against what happened	01-06, 02-06

Figure 9: The watching checklist after Part 2. Line 1 comes from Part 1; lines 2 to 6 are the pre-snap reads Part 2 taught, in the order they become visible during a snap; line 7 is the habit from *How to Watch a Broadcast*. The motion line is this chapter's. The card grows after Parts 4, 6 and 7, and becomes the master checklist of the capstone.

Two habits make the card work. **Say each answer out loud, or write it down, before the snap.** A guess you didn't commit to can't teach you anything. And **read the lines in order.** The motion line comes after the formation line because you can't interpret a defender's reaction until you know what the motion did to the formation; the shell line comes last because the motion may have changed it.

Watch for it on Sunday

💡 Watch for it: follow the motion

For one quarter, every time an offensive player moves before the snap, do four things.

- **Name it.** Jet, orbit, return, zoom, glide, short, a trade, a shift? If you aren't sure, describe the path ("across, in front of the quarterback").
- **Watch who follows.** Pick up the defender who was across from the motion man *before* he moved. Did that defender travel across (T), did the defenders bump over (B), did a safety spin (S)? Write the letters.
- **Call it.** Man or zone? One-high or two-high at the snap? Write your call before the ball is snapped.
- **Note the timing.** Was the ball snapped while he was still moving (motion at the snap), or did he set first?

Then check your calls on the replay, or better, on the All-22 film, which shows the whole secondary. Add a column for it in your prediction log from How to Watch a Broadcast. After a few games you'll notice two things: travelers are rarer than you expect (many man defenses exchange instead), and your calls get better when the motion is at the snap.

Predict the play

💡 Predict the play 1: who went with him?

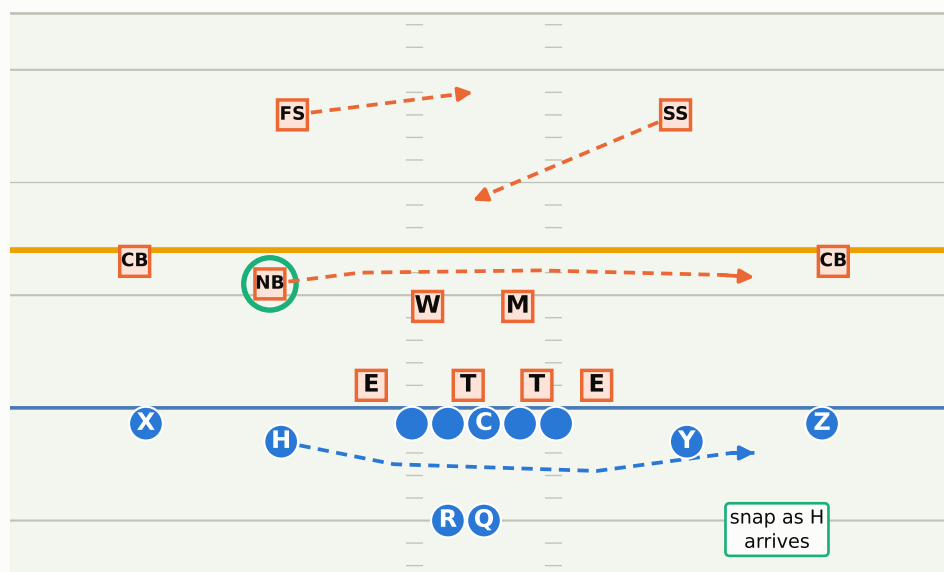


Figure 10: Second-and-7. Before the motion the defense shows two deep safeties. Then H, in the left slot, motions across the formation, and the nickel back (NB, ringed) runs with him the whole way. As he does, the safety on the right (SS) comes down to about 8 yards in the middle of the field and the safety on the left (FS) slides over to the deep middle. Dashed paths are pre-snap movement. The ball is snapped as H reaches the right side.

Man or zone? How many deep safeties? And what does the offense like about the picture?

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play 2: the jet they stopped respecting

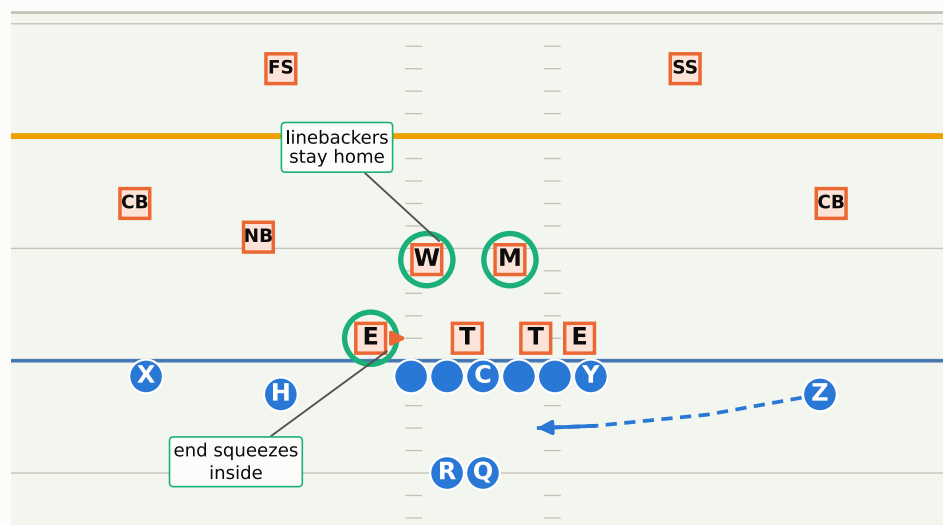


Figure 11: Third quarter, first-and-10. Six times in the first half the offense ran this exact jet motion by Z and handed the ball to the running back going the other way. This time, as Z sprints across, the linebackers (ringed) don't move with him and the defensive end on the side Z is heading to (E, ringed) squeezes inside toward the running back. The safeties stay where they were. Z is 2 yards from the quarterback when the ball is snapped.

What should the offense call here, and why does it work only because of the first half?

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play 3: the back motions out

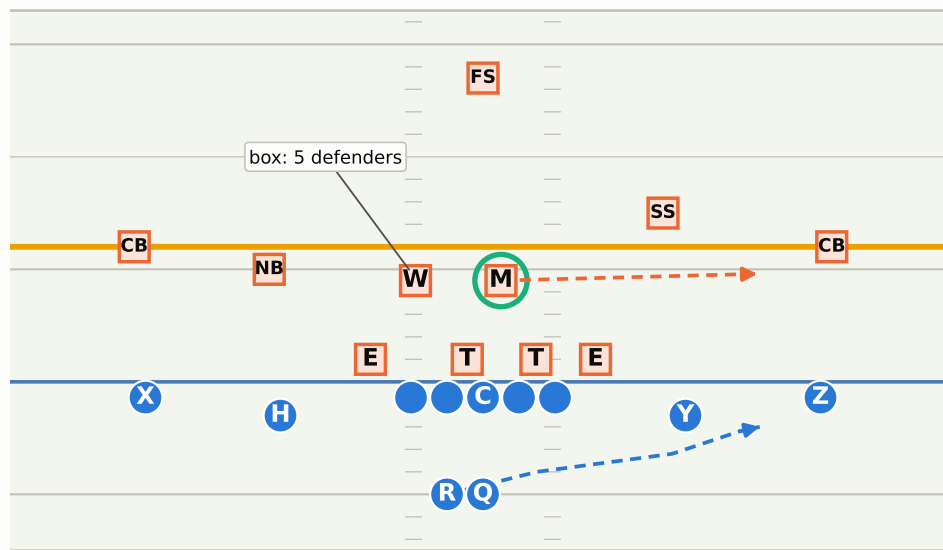


Figure 12: Third-and-6. The running back (R) motions out of the backfield to the right, leaving an empty backfield. The Mike linebacker (M, ringed) walks out of the box with him and stops across from him near the numbers. Everyone else stays put: one deep safety in the middle, the other safety down at 7 yards over the tight end.

Man or zone? And what has the defense just done to its middle?

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- **Motion is never decoration.** Every motion does at least one of five jobs: it asks *man or zone?*, changes the strength and the numbers, creates leverage, gives a running start, or stresses the defense's communication. Every one costs play-clock time and precise timing, and any motion that always means the same play becomes a tell.
- **Learn the shapes, not just the names.** Jet, fly and orbit cross the formation (in front of, behind, and looping behind the quarterback); rocket sprints a back to the edge; return goes and comes back; zoom crosses and sets; glide and short motion come in; a trade or a shift to bunch rebuilds the formation. Names vary by playbook.
- **Read the answer:** a defender who **travels** suggests man; defenders who **bump** suggest zone; safeties who **spin** turn two-high into one-high. That's the **man/zone indicator**, and it is a hint, not a guarantee: man defenses can exchange, pattern-matching zones can follow, and safeties can spin back.
- **The later the motion, the more it's worth.** Motion at the snap gives a running start worth about three yards in the first second, and leaves the defense no time to disguise. NFL use of it rose from about 4% of plays in 2017 to over 20% by 2023 (ESPN), and the league tightened the rules on rolling starts from the line in 2024. American rules allow one man moving sideways or back; Canadian rules allow forward motion.
- **Motion numbers depend on the source.** PFF (any shift or motion: 63.9% in 2025), FTN (motion before or at the snap: 55.1%), ESPN and Next Gen Stats (motion at the snap) all measure different things. Quote one source at a time.
- **The checklist so far:** situation → personnel → formation and strength → the motion response → box → shell, then log it.

Connections

Builds on: The Pre-Snap Phase (the rules of the set, the shift and motion, and the first look at why motion exists), Spread and Modern Formations (2x2, 3x1, bunch, stack and empty, the formations motion moves between), and A First Look at Defense (man and zone, one-high and two-high, the box), with Personnel and Formation Rules and the Language of Alignment behind the checklist's early lines.

Sets up: Perimeter Runs and Misdirection (the jet sweep, the reverse and the plays that live off the jet fake); Coverage Fundamentals and Man Coverage (what travelling and exchanging look like from the defense's side); Disguise, Rotation, and Split-Field Coverage

(how defenses make the motion indicator lie, and the full anatomy of the spin); Play-Action, Bootlegs, and Screens (motion that sells a run); Constraint Theory and Sequencing (why the jet has to be real for the fake to work); and Tendencies: PROE, Personnel, Motion, and Coverage (measuring a team's motion habits yourself). Case studies with motion at their heart: Shanahan's wide-zone tree and Andy Reid's Chiefs.

Go deeper

- **Big Data Bowl 2025** was built around exactly this chapter's question: what pre-snap movement reveals about the play. Its files include, for every player on every play from the first nine weeks of 2022, flags for whether he was in motion at the snap and whether he shifted or went in motion after the line set. Once you have the files, this is how you'd start:

```
import pandas as pd
from gridiron import bdb, show_animation

pp = pd.read_csv("data/bdb/player_play.csv")      # one row per player per play
plays = pd.read_csv("data/bdb/plays.csv")         # one row per play, with PFF's coverage labels
FLAGS = ["inMotionAtBallSnap", "motionSinceLineset", "shiftSinceLineset"]
KEYS = ["gameId", "playId"]
flags = pp.groupby(KEYS)[FLAGS].max().reset_index() # True if any player did it
flags = flags.merge(plays[KEYS + ["possessionTeam", "pff_manZone"]], on=KEYS)
print(flags[FLAGS].mean())                       # share of plays with each
print(pd.crosstab(flags.inMotionAtBallSnap, flags.pff_manZone, normalize="index"))

# animate one play with motion at the snap, drawn like this chapter's diagrams
trk = bdb.load_tracking("data/bdb/tracking_week_1.csv")
g, pl = flags.loc[flags.inMotionAtBallSnap == True, KEYS].iloc[0]
show_animation(bdb.prepare(trk, plays, game_id=g, play_id=pl))
```

- Ben Solak's January 2024 piece in *The Ringer* on how the Chiefs' and Ravens' defenses answered motion, and Nick Wagoner and Lindsey Thiry's 2022 ESPN piece on Shanahan's and McVay's motion habits (both linked in the notes).
- Chris B. Brown, *The Art of Smart Football* (2015), for the older roots of motion and shifts in offensive design.

Answers to Predict the play

Answer 1.

Man coverage, with one deep safety: Cover 1, almost certainly. Two things happened at once. The nickel back travelled with H all the way across the formation, which a zone defender has no reason to do: that says man. And the safeties spun: the two-high picture turned into one safety in the deep middle and one at 8 yards in the middle of the field, a classic spot for a "robber" who lurks underneath looking for crossing routes (see Man Coverage). Travel plus spin is the signature of man coverage disguised as two-high before the motion.

What the offense likes: H's defender is now **behind and inside** a receiver who is moving toward the sideline at full speed. A flat route, a wheel up the sideline, or a quick throw to H the instant the ball is snapped all start as races the nickel back is losing. On the other side, X is alone with his corner, and with the single deep safety in the middle there is no help over the top on the outside. Man coverage also means most defenders have their backs turned to the quarterback, so a quarterback who can run has open grass if the routes clear out the middle.

The caution: a robber in the middle is there precisely to jump the crossing routes that beat man coverage, so throwing a shallow cross into him is the mistake the defense is hoping for.

Answer 2.

Give the ball to Z: the jet sweep itself. The defense has learned that the jet is a fake and the run is coming the other way, so it has stopped spending players on the jet. The linebackers stay home to play the running back, and the end on Z's side is squeezing inside toward the back's path. That leaves the edge on the jet side with nobody setting it except a corner who is seven yards off and blocked by X. Z already has a head start; with the end crashing inside and nobody flowing with him, he gets the corner.

That is the logic of jet motion as a *constraint*: the fake is only worth something if the real thing is a threat. Run the inside handoff enough and the defense ignores the jet; run the jet when it does. Offenses build their game plans around exactly this pairing (more in Perimeter Runs and Misdirection and Constraint Theory and Sequencing). The defense's counter is to keep at least one player honest on the jet side: a safety who spins toward the motion, as in the zone animation earlier, would have been waiting.

Answer 3.

Man coverage. A middle linebacker has no zone responsibility out near the numbers; the only reason for him to walk out with the running back is that the back is *his man*. In zone, the nickel or the safety already standing on that side would simply have taken the back, and the Mike would have stayed in the box.

What it did to the middle: the box now has five defenders (four linemen and the Will), and the middle of the field underneath the deep safety has one fewer player in it. With five receivers out and five blockers, it's also the moment to check the pass protection, because an empty formation leaves the quarterback with no back to pick up an extra rusher; if a sixth defender comes, someone has to be thrown to quickly (see Pass Protection). For the play, man coverage with a linebacker on a running back is a mismatch the offense created on purpose: a quick out or a wheel route by the back, against a linebacker who has to cover him in space, is often the first read. And with every defender's back turned, a quarterback scramble up the middle is open on third-and-6.